



SQL – Joins & Aggregates

SQL - Aggregates

SQL – Aggregate Functions

- ▶ **SQL Aggregate Functions are used to perform calculations on a set of rows and return a single value.**
- ▶ **These functions are particularly useful when we need to summarize, analyze or group large datasets in SQL databases.**

SQL – Aggregate Function

- ▶ **Key Features of Aggregate Functions**
 - ▶ **Operate on groups of rows:** They work on a set of rows and return a single value.
 - ▶ **Ignore NULLs:** Most aggregate functions ignore NULL values, except for COUNT(*).
 - ▶ **Used with GROUP BY:** To perform calculations on grouped data, often use aggregate functions with GROUP BY.
 - ▶ **Can be combined with other SQL clauses:** Aggregate functions can be used alongside HAVING, ORDER BY and other SQL clauses to filter or sort results.

SQL – Aggregate Functions

- ▶ **Commonly Includes:**

- ▶ **COUNT()**

- ▶ **SUM()**

- ▶ **AVG()**

- ▶ **MIN()**

- ▶ **MAX()**

SQL – Aggregates – Count()

- ▶ It is used to count the number of rows in a table. It helps summarize data by giving the total number of entries. It can be used in different ways depending on what you want to count:
- ▶ **COUNT(*):** Counts all rows.
- ▶ **COUNT(column_name):** Counts non-NULL values in the specified column.
- ▶ **COUNT(DISTINCT column_name):** Counts unique non-NULL values in the column.

Examples

-- Total number of records in the table
`SELECT COUNT(*) AS TotalRecords FROM
Employee;`

-- Count of non-NULL salaries
`SELECT COUNT(Salary) AS NonNullSalaries
FROM Employee;`

-- Count of unique non-NULL salaries
`SELECT COUNT(DISTINCT Salary) AS
UniqueSalaries FROM Employee;`

SQL – Aggregates – SUM()

- ▶ It is used to calculate the total of a numeric column. It adds up all non-NULL values in that column
- ▶ SUM(column_name) returns sum of all non-NULL values in the specified column.
- ▶ Example
 - ▶ Calculate the total salary

```
SELECT SUM(Salary) AS TotalSalary FROM  
Employee;
```


SQL – Aggregates – AVG()

- ▶ It is used to calculate average value of a numeric column. It divides sum of all non-NULL values by the number of non-NULL rows.
 - ▶ **AVG(column_name)** returns average of all non-NULL values in the specified column.
 - ▶ Example
 - ▶ Calculate the Average salary
- ```
SELECT AVG(Salary) AS AverageSalary FROM Employee;
```

## SQL – Aggregates – MIN()

- ▶ It is used to find the minimum non-null value of a numeric column.
  - ▶ **MIN(column\_name)** returns the lowest non-null value of the specified column.
  - ▶ Example
    - ▶ Finding the lowest salary
- ```
SELECT MIN(Salary) AS MinimumSalary FROM Employee;
```

SQL – Aggregates – MAX()

- ▶ It is used to find the maximum non-null value of a numeric column.
 - ▶ **MAX(column_name)** returns the highest non-null value of the specified column.
 - ▶ Example
 - ▶ Finding the Highest salary
- ```
SELECT MAX(Salary) AS MaximumSalary FROM Employee;
```

# SQL - Joins



# SQL – Joins

- ▶ **SQL joins are fundamental tools for combining data from multiple tables in relational databases.**
- ▶ **Real-world Scenarios:**
  - ▶ **Students and their Marks / grades**
  - ▶ **Employees and their Salaries**
  - ▶ **Departments and their Employees**
  - ▶ **Customer Orders including Products from Suppliers / Vendors**

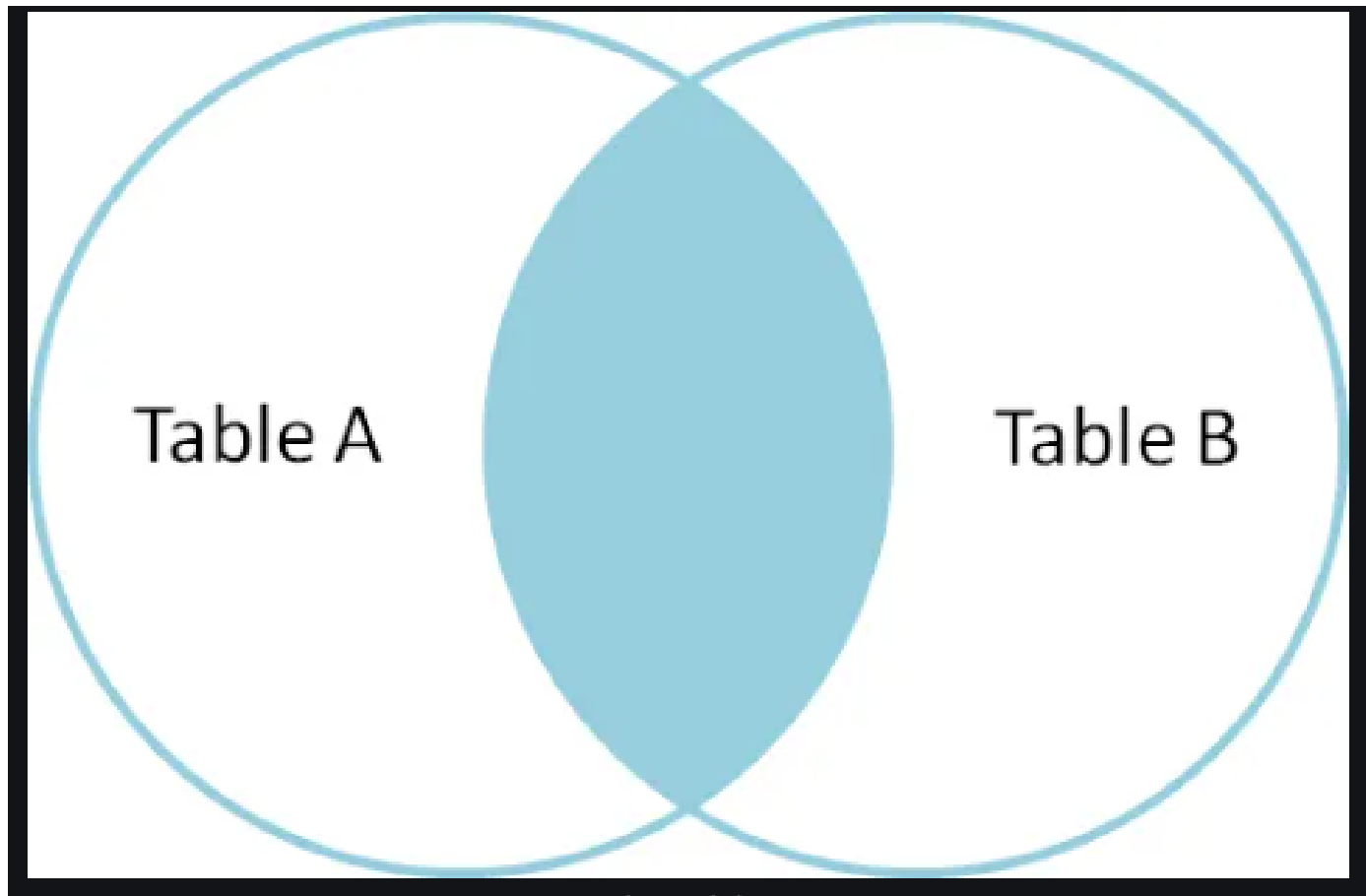
# SQL – Joins - Types

- ▶ **Inner Join**
- ▶ **Natural Join**
- ▶ **Outer Join**

# SQL – Joins - Syntax

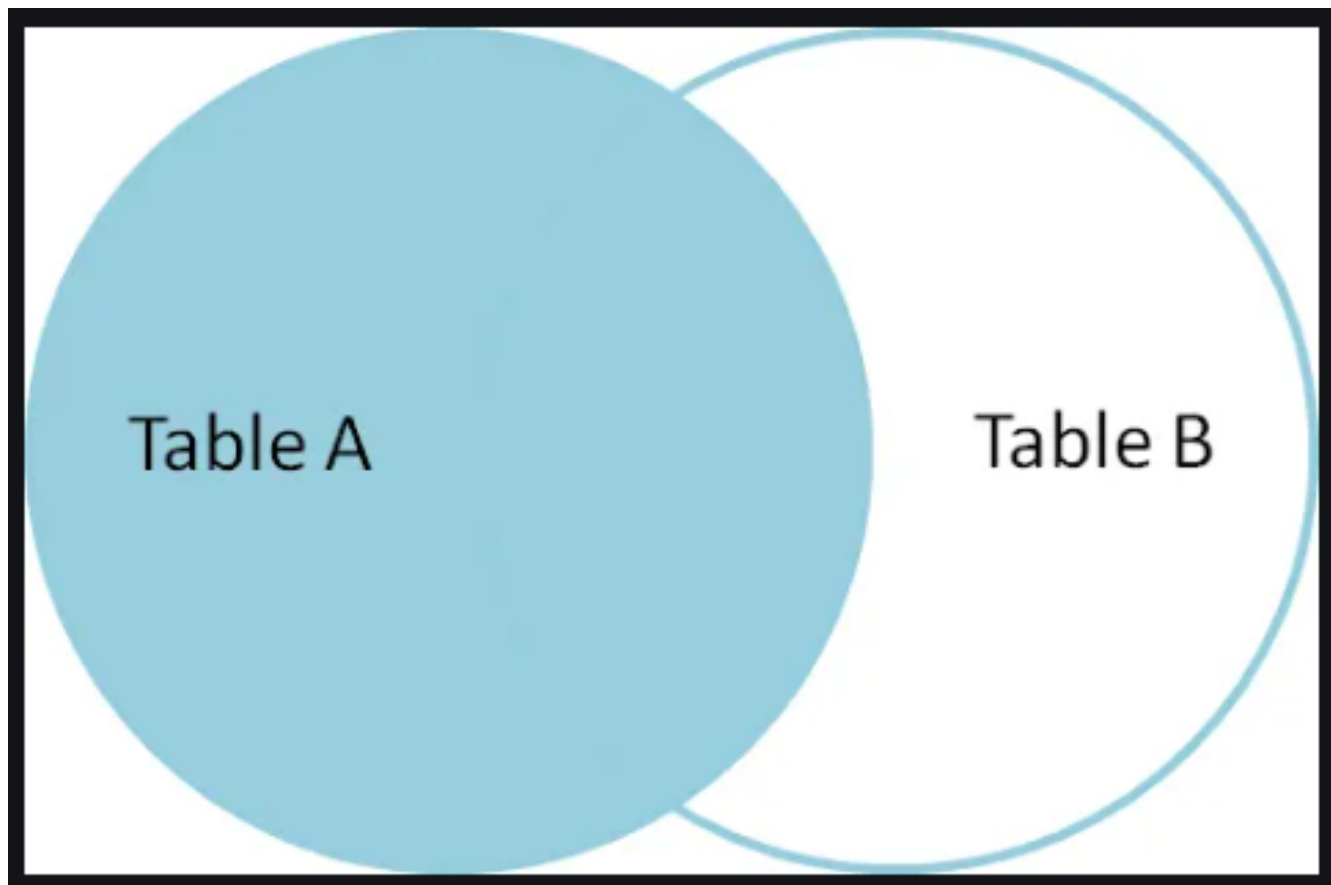
- ▶ **SELECT**  
table1.column1,table1.column2,table2.column1,....
- ▶ **FROM** table1 **INNER JOIN** table2
- ▶ **ON** table1.matchingcolumn =  
table2.matchingcolumn

# SQL – Joins – Inner Join / Join

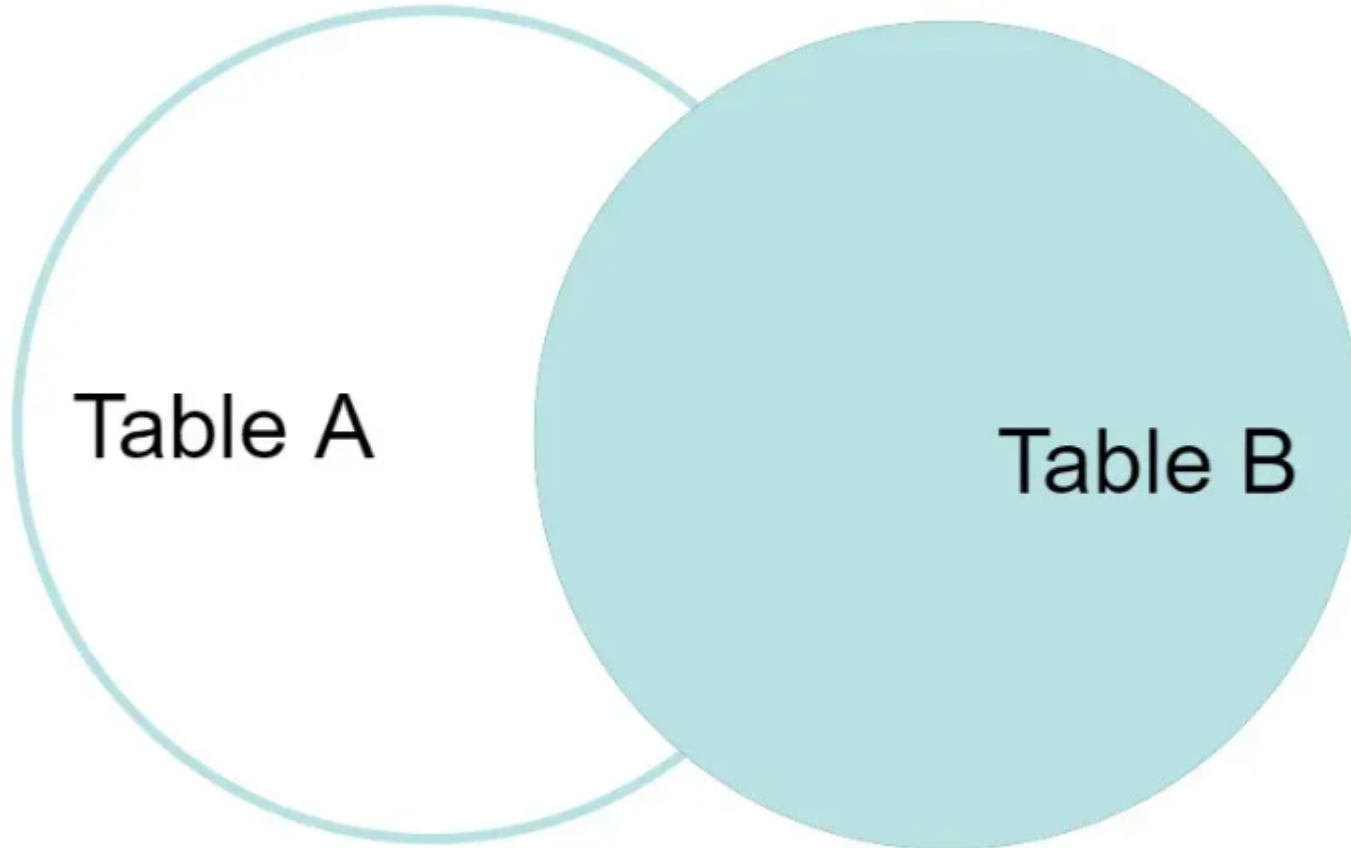




# SQL – Joins – Left Outer Join



# SQL – Joins – Right Outer Join



# SQL – Joins – Natural Join

- ▶ A **Natural Join** in SQL is a type of **join operation** that automatically combines tables based on **columns with the same names and compatible data types** in both tables.

**SELECT \***

**FROM table1**

**NATURAL JOIN table2;**

# Example

| StudentID | Name  | DeptID |
|-----------|-------|--------|
| 1         | Ali   | D1     |
| 2         | Sara  | D2     |
| 3         | Bilal | D1     |
| 4         | Hina  | D3     |
| 5         | Ahmed | D4     |

| DeptID | DeptName         |
|--------|------------------|
| D1     | Computer Science |
| D2     | Business Admin   |
| D3     | Electrical Engg  |
| D4     | Psychology       |
| D5     | Architecture     |

# INNER JOIN

- ▶ Combines the two relations with only relevant tuples, based on the predicate specified in the ***where*** clause
- ▶ Operation used in SQL:
  - ▶ **join**
  - ▶ **inner join**
- ▶ The predicate can be specified by keywords: ***on*** or ***where***
- ▶ Both ***on*** and ***where*** work the same way for inner join, but different for outer joins



# INNER JOIN

► Example:

► **select \* from instructor join teaches on instructor.ID = teaches.ID;**

| ID    | name       | dept_name  | salary |
|-------|------------|------------|--------|
| 10101 | Srinivasan | Comp. Sci. | 65000  |
| 12121 | Wu         | Finance    | 90000  |
| 15151 | Mozart     | Music      | 40000  |
| 22222 | Einstein   | Physics    | 95000  |
| 32343 | El Said    | History    | 60000  |
| 33456 | Gold       | Physics    | 87000  |
| 45565 | Katz       | Comp. Sci. | 75000  |
| 58583 | Califieri  | History    | 62000  |
| 76543 | Singh      | Finance    | 80000  |
| 76766 | Crick      | Biology    | 72000  |
| 83821 | Brandt     | Comp. Sci. | 92000  |
| 98345 | Kim        | Elec. Eng. | 80000  |

The instructor relation

| ID    | course_id | sec_id | semester | year |
|-------|-----------|--------|----------|------|
| 10101 | CS-101    | 1      | Fall     | 2017 |
| 10101 | CS-315    | 1      | Spring   | 2018 |
| 10101 | CS-347    | 1      | Fall     | 2017 |
| 12121 | FIN-201   | 1      | Spring   | 2018 |
| 15151 | MU-199    | 1      | Spring   | 2018 |
| 22222 | PHY-101   | 1      | Fall     | 2017 |
| 32343 | HIS-351   | 1      | Spring   | 2018 |
| 45565 | CS-101    | 1      | Spring   | 2018 |
| 45565 | CS-319    | 1      | Spring   | 2018 |
| 76766 | BIO-101   | 1      | Summer   | 2017 |
| 76766 | BIO-301   | 1      | Summer   | 2018 |
| 83821 | CS-190    | 1      | Spring   | 2017 |
| 83821 | CS-190    | 2      | Spring   | 2017 |
| 83821 | CS-319    | 2      | Spring   | 2018 |
| 98345 | EE-181    | 1      | Spring   | 2017 |

The teaches relation

| instructor.ID | name       | dept_name  | salary | teaches.ID | course_id | sec_id | semester | year |
|---------------|------------|------------|--------|------------|-----------|--------|----------|------|
| 10101         | Srinivasan | Comp. Sci. | 65000  | 10101      | CS-101    | 1      | Fall     | 2017 |
| 10101         | Srinivasan | Comp. Sci. | 65000  | 10101      | CS-315    | 1      | Spring   | 2018 |
| 10101         | Srinivasan | Comp. Sci. | 65000  | 10101      | CS-347    | 1      | Fall     | 2017 |
| 12121         | Wu         | Finance    | 90000  | 12121      | FIN-201   | 1      | Spring   | 2018 |
| 15151         | Mozart     | Music      | 40000  | 15151      | MU-199    | 1      | Spring   | 2018 |
| 22222         | Einstein   | Physics    | 95000  | 22222      | PHY-101   | 1      | Fall     | 2017 |
| 32343         | El Said    | History    | 60000  | 32343      | HIS-351   | 1      | Spring   | 2018 |
| 45565         | Katz       | Comp. Sci. | 75000  | 45565      | CS-101    | 1      | Spring   | 2018 |
| 45565         | Katz       | Comp. Sci. | 75000  | 45565      | CS-319    | 1      | Spring   | 2018 |
| 76766         | Crick      | Biology    | 72000  | 76766      | BIO-101   | 1      | Summer   | 2017 |
| 76766         | Crick      | Biology    | 72000  | 76766      | BIO-301   | 1      | Summer   | 2018 |
| 83821         | Brandt     | Comp. Sci. | 92000  | 83821      | CS-190    | 1      | Spring   | 2017 |
| 83821         | Brandt     | Comp. Sci. | 92000  | 83821      | CS-190    | 2      | Spring   | 2017 |
| 83821         | Brandt     | Comp. Sci. | 92000  | 83821      | CS-319    | 2      | Spring   | 2018 |
| 98345         | Kim        | Elec. Eng. | 80000  | 98345      | EE-181    | 1      | Spring   | 2017 |

# INNER JOIN

- Can explicitly use *inner join* to do the same.
- For example

| ID    | name       | dept_name  | salary |
|-------|------------|------------|--------|
| 10101 | Srinivasan | Comp. Sci. | 65000  |
| 12121 | Wu         | Finance    | 90000  |
| 15151 | Mozart     | Music      | 40000  |
| 22222 | Einstein   | Physics    | 95000  |
| 32343 | El Said    | History    | 60000  |
| 33456 | Gold       | Physics    | 87000  |
| 45565 | Katz       | Comp. Sci. | 75000  |
| 58583 | Califieri  | History    | 62000  |
| 76543 | Singh      | Finance    | 80000  |
| 76766 | Crick      | Biology    | 72000  |
| 83821 | Brandt     | Comp. Sci. | 92000  |
| 98345 | Kim        | Elec. Eng. | 80000  |

The *instructor* relation

| ID    | course_id | sec_id | semester | year |
|-------|-----------|--------|----------|------|
| 10101 | CS-101    | 1      | Fall     | 2017 |
| 10101 | CS-315    | 1      | Spring   | 2018 |
| 10101 | CS-347    | 1      | Fall     | 2017 |
| 12121 | FIN-201   | 1      | Spring   | 2018 |
| 15151 | MU-199    | 1      | Spring   | 2018 |
| 22222 | PHY-101   | 1      | Fall     | 2017 |
| 32343 | HIS-351   | 1      | Spring   | 2018 |
| 45565 | CS-101    | 1      | Spring   | 2018 |
| 45565 | CS-319    | 1      | Spring   | 2018 |
| 76766 | BIO-101   | 1      | Summer   | 2017 |
| 76766 | BIO-301   | 1      | Summer   | 2018 |
| 83821 | CS-190    | 1      | Spring   | 2017 |
| 83821 | CS-190    | 2      | Spring   | 2017 |
| 83821 | CS-319    | 2      | Spring   | 2018 |
| 98345 | EE-181    | 1      | Spring   | 2017 |

The *teaches* relation

inner join

| instructor.ID | name       | dept_name  | salary | teaches.ID | course_id | sec_id | semester | year |
|---------------|------------|------------|--------|------------|-----------|--------|----------|------|
| 10101         | Srinivasan | Comp. Sci. | 65000  | 10101      | CS-101    | 1      | Fall     | 2017 |
| 10101         | Srinivasan | Comp. Sci. | 65000  | 10101      | CS-315    | 1      | Spring   | 2018 |
| 10101         | Srinivasan | Comp. Sci. | 65000  | 10101      | CS-347    | 1      | Fall     | 2017 |
| 12121         | Wu         | Finance    | 90000  | 12121      | FIN-201   | 1      | Spring   | 2018 |
| 15151         | Mozart     | Music      | 40000  | 15151      | MU-199    | 1      | Spring   | 2018 |
| 22222         | Einstein   | Physics    | 95000  | 22222      | PHY-101   | 1      | Fall     | 2017 |
| 32343         | El Said    | History    | 60000  | 32343      | HIS-351   | 1      | Spring   | 2018 |
| 45565         | Katz       | Comp. Sci. | 75000  | 45565      | CS-101    | 1      | Spring   | 2018 |
| 45565         | Katz       | Comp. Sci. | 75000  | 45565      | CS-319    | 1      | Spring   | 2018 |
| 76766         | Crick      | Biology    | 72000  | 76766      | BIO-101   | 1      | Summer   | 2017 |
| 76766         | Crick      | Biology    | 72000  | 76766      | BIO-301   | 1      | Summer   | 2018 |
| 83821         | Brandt     | Comp. Sci. | 92000  | 83821      | CS-190    | 1      | Spring   | 2017 |
| 83821         | Brandt     | Comp. Sci. | 92000  | 83821      | CS-190    | 2      | Spring   | 2017 |
| 83821         | Brandt     | Comp. Sci. | 92000  | 83821      | CS-319    | 2      | Spring   | 2018 |
| 98345         | Kim        | Elec. Eng. | 80000  | 98345      | EE-181    | 1      | Spring   | 2017 |

# NATURAL JOIN

- ▶ Same as inner join, except it removes the redundant attributes from the final table
- ▶ Does not need predicates, because it works on same attributes by default.
- ▶ May use ***where*** clause if some other condition apart from equality needs to be checked
- ▶ In SQL use ***natural join*** in queries

# NATURAL JOIN

► Example:

► **select \*** from *instructor* natural join *teaches*;

| ID    | name       | dept_name  | salary |
|-------|------------|------------|--------|
| 10101 | Srinivasan | Comp. Sci. | 65000  |
| 12121 | Wu         | Finance    | 90000  |
| 15151 | Mozart     | Music      | 40000  |
| 22222 | Einstein   | Physics    | 95000  |
| 32343 | El Said    | History    | 60000  |
| 33456 | Gold       | Physics    | 87000  |
| 45565 | Katz       | Comp. Sci. | 75000  |
| 58583 | Califleri  | History    | 62000  |
| 76543 | Singh      | Finance    | 80000  |
| 76766 | Crick      | Biology    | 72000  |
| 83821 | Brandt     | Comp. Sci. | 92000  |
| 98345 | Kim        | Elec. Eng. | 80000  |

The *instructor* relation

| ID    | course_id | sec_id | semester | year |
|-------|-----------|--------|----------|------|
| 10101 | CS-101    | 1      | Fall     | 2017 |
| 10101 | CS-315    | 1      | Spring   | 2018 |
| 10101 | CS-347    | 1      | Fall     | 2017 |
| 12121 | FIN-201   | 1      | Spring   | 2018 |
| 15151 | MU-199    | 1      | Spring   | 2018 |
| 22222 | PHY-101   | 1      | Fall     | 2017 |
| 32343 | HIS-351   | 1      | Spring   | 2018 |
| 45565 | CS-101    | 1      | Spring   | 2018 |
| 45565 | CS-319    | 1      | Spring   | 2018 |
| 76766 | BIO-101   | 1      | Summer   | 2017 |
| 76766 | BIO-301   | 1      | Summer   | 2018 |
| 83821 | CS-190    | 1      | Spring   | 2017 |
| 83821 | CS-190    | 2      | Spring   | 2017 |
| 83821 | CS-319    | 2      | Spring   | 2018 |
| 98345 | EE-181    | 1      | Spring   | 2017 |

The *teaches* relation

| ID    | name       | dept_name  | salary | course_id | sec_id | semester | year |
|-------|------------|------------|--------|-----------|--------|----------|------|
| 10101 | Srinivasan | Comp. Sci. | 65000  | CS-101    | 1      | Fall     | 2017 |
| 10101 | Srinivasan | Comp. Sci. | 65000  | CS-315    | 1      | Spring   | 2018 |
| 10101 | Srinivasan | Comp. Sci. | 65000  | CS-347    | 1      | Fall     | 2017 |
| 12121 | Wu         | Finance    | 90000  | FIN-201   | 1      | Spring   | 2018 |
| 15151 | Mozart     | Music      | 40000  | MU-199    | 1      | Spring   | 2018 |
| 22222 | Einstein   | Physics    | 95000  | PHY-101   | 1      | Fall     | 2017 |
| 32343 | El Said    | History    | 60000  | HIS-351   | 1      | Spring   | 2018 |
| 45565 | Katz       | Comp. Sci. | 75000  | CS-101    | 1      | Spring   | 2018 |
| 45565 | Katz       | Comp. Sci. | 75000  | CS-319    | 1      | Spring   | 2018 |
| 76766 | Crick      | Biology    | 72000  | BIO-101   | 1      | Summer   | 2017 |
| 76766 | Crick      | Biology    | 72000  | BIO-301   | 1      | Summer   | 2018 |
| 83821 | Brandt     | Comp. Sci. | 92000  | CS-190    | 1      | Spring   | 2017 |
| 83821 | Brandt     | Comp. Sci. | 92000  | CS-190    | 2      | Spring   | 2017 |
| 83821 | Brandt     | Comp. Sci. | 92000  | CS-319    | 2      | Spring   | 2018 |
| 98345 | Kim        | Elec. Eng. | 80000  | EE-181    | 1      | Spring   | 2017 |

# NATURAL JOIN

- ▶ Can combine natural join with Cartesian Product
- ▶ Example:
  - ▶ **select** \*  
**from** *instructor* **natural join** *teaches*, *course*  
**where** *teaches*.courseID = *course*.courseID;
  - ▶ This joins the relations: *instructor* and *teaches* based on ID first
  - ▶ Then use this resultant relation with *course* and filter the result based on course ID



# OUTER JOIN

- ▶ An outer join is used where the user wants to keep all the tuples in  $R$ , or all those in  $S$ , or all those in both relations in the result of the JOIN, regardless of whether or not they have matching tuples in the other relation.
- ▶ A **left outer join** operation keeps every tuple in the first, or left, relation  $R$  in  $R \bowtie S$ ; if no matching tuple is found in  $S$ , then the attributes of  $S$  in the join result are filled or padded with NULL values.
- ▶ In SQL use ***left outer join*** or ***natural left outer join*** in queries

# LEFT OUTER JOIN (EXAMPLE)

- For example:
  - select** *Fname, Minit, Lname, Dname* **from** *Employee* **natural left outer join** *Department*

EMPLOYEE

| Fname    | Minit | Lname   | Ssn       | Dno |
|----------|-------|---------|-----------|-----|
| John     | B     | Smith   | 123456789 | 3   |
| Franklin | T     | Wong    | 333445555 | 5   |
| Alicia   | J     | Zelaya  | 999887777 | 3   |
| Jennifer | S     | Wallace | 987654321 | 4   |
| Ramesh   | K     | Narayan | 666884444 | 3   |
| Joyce    | A     | English | 453453453 | 1   |
| Ahmad    | V     | Jabbar  | 987987987 | 4   |
| James    | E     | Borg    | 888665555 | 1   |

DEPARTMENT

| Dname          | Dnumber | Mgr_ssn   | Mgr_start_date |
|----------------|---------|-----------|----------------|
| Research       | 5       | 333445555 | 1988-05-22     |
| Administration | 4       | 987654321 | 1995-01-01     |
| Headquarters   | 1       | 888665555 | 1981-06-19     |

RESULT

| Fname    | Minit | Lname   | Dname          |
|----------|-------|---------|----------------|
| John     | B     | Smith   | NULL           |
| Franklin | T     | Wong    | Research       |
| Alicia   | J     | Zelaya  | NULL           |
| Jennifer | S     | Wallace | Administration |
| Ramesh   | K     | Narayan | NULL           |
| Joyce    | A     | English | Headquarters   |
| Ahmad    | V     | Jabbar  | Administration |
| James    | E     | Borg    | Headquarters   |

# LEFT OUTER JOIN

► For example:

| ID    | name     | dept_name  | tot_cred |
|-------|----------|------------|----------|
| 00128 | Zhang    | Comp. Sci. | 102      |
| 12345 | Shankar  | Comp. Sci. | 32       |
| 19991 | Brandt   | History    | 80       |
| 23121 | Chavez   | Finance    | 110      |
| 44553 | Peltier  | Physics    | 56       |
| 45678 | Levy     | Physics    | 46       |
| 54321 | Williams | Comp. Sci. | 54       |
| 55739 | Sanchez  | Music      | 38       |
| 70557 | Snow     | Physics    | 0        |
| 76543 | Brown    | Comp. Sci. | 58       |
| 76653 | Aoi      | Elec. Eng. | 60       |
| 98765 | Bourikas | Elec. Eng. | 98       |
| 98988 | Tanaka   | Biology    | 120      |

Figure 4.1 The *student* relation.

| ID    | course_id | sec_id | semester | year | grade |
|-------|-----------|--------|----------|------|-------|
| 00128 | CS-101    | 1      | Fall     | 2017 | A     |
| 00128 | CS-347    | 1      | Fall     | 2017 | A-    |
| 12345 | CS-101    | 1      | Fall     | 2017 | C     |
| 12345 | CS-190    | 2      | Spring   | 2017 | A     |
| 12345 | CS-315    | 1      | Spring   | 2018 | A     |
| 12345 | CS-347    | 1      | Fall     | 2017 | A     |
| 19991 | HIS-351   | 1      | Spring   | 2018 | B     |
| 23121 | FIN-201   | 1      | Spring   | 2018 | C+    |
| 44553 | PHY-101   | 1      | Fall     | 2017 | B-    |
| 45678 | CS-101    | 1      | Fall     | 2017 | F     |
| 45678 | CS-101    | 1      | Spring   | 2018 | B+    |
| 45678 | CS-319    | 1      | Spring   | 2018 | B     |
| 54321 | CS-101    | 1      | Fall     | 2017 | A-    |
| 54321 | CS-190    | 2      | Spring   | 2017 | B+    |
| 55739 | MU-199    | 1      | Spring   | 2018 | A-    |
| 76543 | CS-101    | 1      | Fall     | 2017 | A     |
| 76543 | CS-319    | 2      | Spring   | 2018 | A     |
| 76653 | EE-181    | 1      | Spring   | 2017 | C     |
| 98765 | CS-101    | 1      | Fall     | 2017 | C-    |
| 98765 | CS-315    | 1      | Spring   | 2018 | B     |
| 98988 | BIO-101   | 1      | Summer   | 2017 | A     |
| 98988 | BIO-301   | 1      | Summer   | 2018 | null  |

Figure 4.2 The *takes* relation.

takes c

| ID    | name     | dept_name  | tot_cred | course_id | sec_id | semester | year | grade |
|-------|----------|------------|----------|-----------|--------|----------|------|-------|
| 00128 | Zhang    | Comp. Sci. | 102      | CS-101    | 1      | Fall     | 2017 | A     |
| 00128 | Zhang    | Comp. Sci. | 102      | CS-347    | 1      | Fall     | 2017 | A-    |
| 12345 | Shankar  | Comp. Sci. | 32       | CS-101    | 1      | Fall     | 2017 | C     |
| 12345 | Shankar  | Comp. Sci. | 32       | CS-190    | 2      | Spring   | 2017 | A     |
| 12345 | Shankar  | Comp. Sci. | 32       | CS-315    | 1      | Spring   | 2018 | A     |
| 12345 | Shankar  | Comp. Sci. | 32       | CS-347    | 1      | Fall     | 2017 | A     |
| 19991 | Brandt   | History    | 80       | HIS-351   | 1      | Spring   | 2018 | B     |
| 23121 | Chavez   | Finance    | 110      | FIN-201   | 1      | Spring   | 2018 | C+    |
| 44553 | Peltier  | Physics    | 56       | PHY-101   | 1      | Fall     | 2017 | B-    |
| 45678 | Levy     | Physics    | 46       | CS-101    | 1      | Fall     | 2017 | F     |
| 45678 | Levy     | Physics    | 46       | CS-101    | 1      | Spring   | 2018 | B+    |
| 45678 | Levy     | Physics    | 46       | CS-319    | 1      | Spring   | 2018 | B     |
| 54321 | Williams | Comp. Sci. | 54       | CS-101    | 1      | Fall     | 2017 | A-    |
| 54321 | Williams | Comp. Sci. | 54       | CS-190    | 2      | Spring   | 2017 | B+    |
| 55739 | Sanchez  | Music      | 38       | MU-199    | 1      | Spring   | 2018 | A-    |
| 70557 | Snow     | Physics    | 0        | null      | null   | null     | null | null  |
| 76543 | Brown    | Comp. Sci. | 58       | CS-101    | 1      | Fall     | 2017 | A     |
| 76543 | Brown    | Comp. Sci. | 58       | CS-319    | 2      | Spring   | 2018 | A     |
| 76653 | Aoi      | Elec. Eng. | 60       | EE-181    | 1      | Spring   | 2017 | C     |
| 98765 | Bourikas | Elec. Eng. | 98       | CS-101    | 1      | Fall     | 2017 | C-    |
| 98765 | Bourikas | Elec. Eng. | 98       | CS-315    | 1      | Spring   | 2018 | B     |
| 98988 | Tanaka   | Biology    | 120      | BIO-101   | 1      | Summer   | 2017 | A     |
| 98988 | Tanaka   | Biology    | 120      | BIO-301   | 1      | Summer   | 2018 | null  |

Figure 4.4 Result of *student* natural left outer join *takes*.

# RIGHT OUTER JOIN

- ▶ A similar operation **right outer join** ( $\bowtie$ ), keeps every tuple in the second, or right, relation  $S$  in the result of  $R \bowtie S$ .
- ▶ In SQL use *right outer join* or *natural right outer join* in queries

# RIGHT OUTER JOIN

► For example:

| ID    | name     | dept_name  | tot_cred |
|-------|----------|------------|----------|
| 00128 | Zhang    | Comp. Sci. | 102      |
| 12345 | Shankar  | Comp. Sci. | 32       |
| 19991 | Brandt   | History    | 80       |
| 23121 | Chavez   | Finance    | 110      |
| 44553 | Peltier  | Physics    | 56       |
| 45678 | Levy     | Physics    | 46       |
| 54321 | Williams | Comp. Sci. | 54       |
| 55739 | Sanchez  | Music      | 38       |
| 70557 | Snow     | Physics    | 0        |
| 76543 | Brown    | Comp. Sci. | 58       |
| 76653 | Aoi      | Elec. Eng. | 60       |
| 98765 | Bourikas | Elec. Eng. | 98       |
| 98988 | Tanaka   | Biology    | 120      |

Figure 4.1 The *student* relation.

| ID    | course_id | sec_id | semester | year | grade |
|-------|-----------|--------|----------|------|-------|
| 00128 | CS-101    | 1      | Fall     | 2017 | A     |
| 00128 | CS-347    | 1      | Fall     | 2017 | A-    |
| 12345 | CS-101    | 1      | Fall     | 2017 | C     |
| 12345 | CS-190    | 2      | Spring   | 2017 | A     |
| 12345 | CS-315    | 1      | Spring   | 2018 | A     |
| 12345 | CS-347    | 1      | Fall     | 2017 | A     |
| 19991 | HIS-351   | 1      | Spring   | 2018 | B     |
| 23121 | FIN-201   | 1      | Spring   | 2018 | C+    |
| 44553 | PHY-101   | 1      | Fall     | 2017 | B-    |
| 45678 | CS-101    | 1      | Fall     | 2017 | F     |
| 45678 | CS-101    | 1      | Spring   | 2018 | B+    |
| 45678 | CS-319    | 1      | Spring   | 2018 | B     |
| 54321 | CS-101    | 1      | Fall     | 2017 | A-    |
| 54321 | CS-190    | 2      | Spring   | 2017 | B+    |
| 55739 | MU-199    | 1      | Spring   | 2018 | A-    |
| 76543 | CS-101    | 1      | Fall     | 2017 | A     |
| 76543 | CS-319    | 2      | Spring   | 2018 | A     |
| 76653 | EE-181    | 1      | Spring   | 2017 | C     |
| 98765 | CS-101    | 1      | Fall     | 2017 | C-    |
| 98765 | CS-315    | 1      | Spring   | 2018 | B     |
| 98988 | BIO-101   | 1      | Summer   | 2017 | A     |
| 98988 | BIO-301   | 1      | Summer   | 2018 | null  |

Figure 4.2 The *takes* relation.

| ID    | course_id | sec_id | semester | year | grade | name     | dept_name  | tot_cred |
|-------|-----------|--------|----------|------|-------|----------|------------|----------|
| 00128 | CS-101    | 1      | Fall     | 2017 | A     | Zhang    | Comp. Sci. | 102      |
| 00128 | CS-347    | 1      | Fall     | 2017 | A-    | Zhang    | Comp. Sci. | 102      |
| 12345 | CS-101    | 1      | Fall     | 2017 | C     | Shankar  | Comp. Sci. | 32       |
| 12345 | CS-190    | 2      | Spring   | 2017 | A     | Shankar  | Comp. Sci. | 32       |
| 12345 | CS-315    | 1      | Spring   | 2018 | A     | Shankar  | Comp. Sci. | 32       |
| 12345 | CS-347    | 1      | Fall     | 2017 | A     | Shankar  | Comp. Sci. | 32       |
| 19991 | HIS-351   | 1      | Spring   | 2018 | B     | Brandt   | History    | 80       |
| 23121 | FIN-201   | 1      | Spring   | 2018 | C+    | Chavez   | Finance    | 110      |
| 44553 | PHY-101   | 1      | Fall     | 2017 | B-    | Peltier  | Physics    | 56       |
| 45678 | CS-101    | 1      | Fall     | 2017 | F     | Levy     | Physics    | 46       |
| 45678 | CS-101    | 1      | Spring   | 2018 | B+    | Levy     | Physics    | 46       |
| 45678 | CS-319    | 1      | Spring   | 2018 | B     | Levy     | Physics    | 46       |
| 54321 | CS-101    | 1      | Fall     | 2017 | A-    | Williams | Comp. Sci. | 54       |
| 54321 | CS-190    | 2      | Spring   | 2017 | B+    | Williams | Comp. Sci. | 54       |
| 55739 | MU-199    | 1      | Spring   | 2018 | A-    | Sanchez  | Music      | 38       |
| 70557 | null      | null   | null     | null | null  | Snow     | Physics    | 0        |
| 76543 | CS-101    | 1      | Fall     | 2017 | A     | Brown    | Comp. Sci. | 58       |
| 76543 | CS-319    | 2      | Spring   | 2018 | A     | Brown    | Comp. Sci. | 58       |
| 76653 | EE-181    | 1      | Spring   | 2017 | C     | Aoi      | Elec. Eng. | 60       |
| 98765 | CS-101    | 1      | Fall     | 2017 | C-    | Bourikas | Elec. Eng. | 98       |
| 98765 | CS-315    | 1      | Spring   | 2018 | B     | Bourikas | Elec. Eng. | 98       |
| 98988 | BIO-101   | 1      | Summer   | 2017 | A     | Tanaka   | Biology    | 120      |
| 98988 | BIO-301   | 1      | Summer   | 2018 | null  | Tanaka   | Biology    | 120      |

Figure 4.5 The result of *takes* natural right outer join *student*.



# FULL OUTER JOIN

- ▶ A **full outer join** denoted by  $\bowtie$ , keeps all tuples in both the left and the right relations when no matching tuples are found, padding them with NULL values as needed.
- ▶ In SQL use ***full outer join*** or ***natural full outer join*** in queries
- ▶ In case of ***on*** and ***where*** there is a difference
  - ▶ ***On*** is applied on join
  - ▶ ***Where*** is applied on resultant tuples



THANK YOU